QUANTUM ERROR CORRECTION (QEC) — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: April 27, 2026

Diagram-based questions on Quantum Error Correction (QEC)

Question 1

10.0 points

Analyze the syndrome measurement circuit for a 3-qubit bit-flip code as shown in the circuit below. Explain how the circuit detects a single-bit error and what each ancilla qubit measurement indicates about the error location.

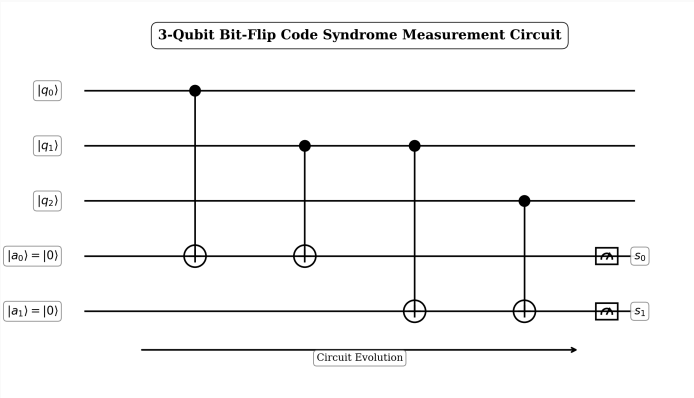

Fig. 1. Quantum Circuit

Question 2

10.0 points

Interpret the lattice structure of the surface code and identify the logical qubits as represented in the diagram below. Discuss the significance of the code distance in determining the fault tolerance.

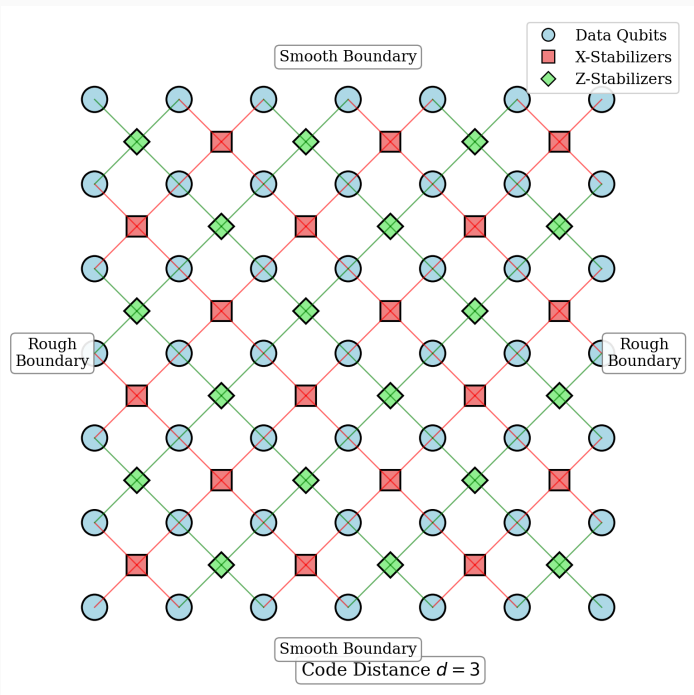

Fig. 2. Surface Code Lattice

Question 3

10.0 points

Analyze the diagram below showing fault-tolerant gate teleportation. Describe the process involved and how it ensures error correction while executing gates.

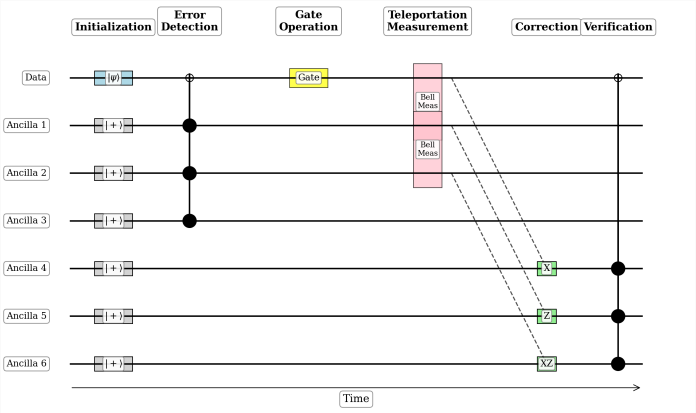

Fig. 3. Quantum Circuit

Question 4

10.0 points

Examine the error-rate threshold theorem diagram below. Explain how different regions represent the operational boundaries of quantum error correction and its implications for scalability.

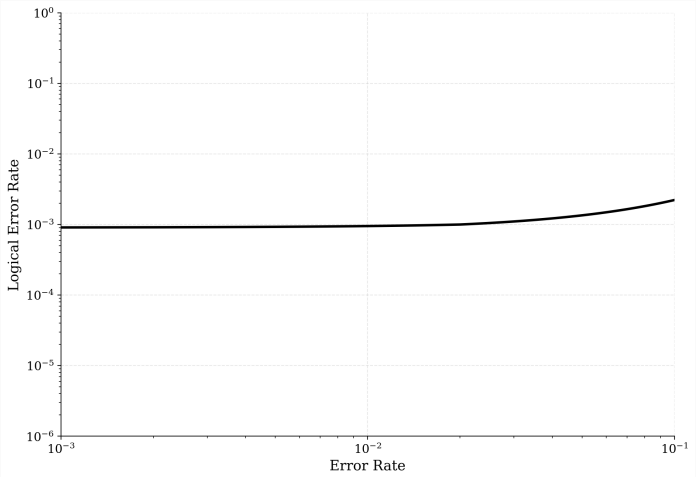

Fig. 4. Error Rate Threshold Diagram

Question 5 10.0 points

Describe the role of stabilizer codes in quantum error correction and how they form the foundation of error detection as illustrated in the syndrome extraction process in the diagram below.

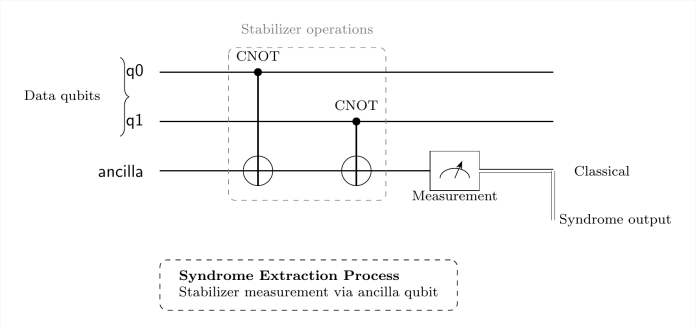

Fig. 5. Syndrome Extraction Process